Analysis / polylogarithms / transseries

Resonant Polylogarithm Inversion

Uniform Order Crossover, Gamma–Zeta Fold Loci, and Convergent Exponential Sectors

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11 October 2026

Abstract

The familiar expansion of a polylogarithm at its unit argument is not a uniform inverse expansion when its order approaches an integer. We retain the cancelling Gamma and zeta poles as one holomorphic block, derive every integer-order logarithmic jet, and construct a convergent inverse chart in three small variables. The chart's crossover function is $\mathcal{A}(\lambda) = (1 - e^{-\lambda})/\lambda$; its nonzero complex zeros require a different scaling.

At these zeros we obtain an exact Gamma–zeta formula for the critical locus of the singular core at every integer order, and compute the first analytic-tail deformation of that locus. The dilogarithm is exceptional: its undetuned secondary resonance is already a fold. If $y = e^{-X}$, $\lambda_k = 2\pi ik \neq 0$, and $\varepsilon = \lambda_k/X$, the equation $\zeta(2+\varepsilon) - \mathrm{Li}_{2+\varepsilon}(e^{-t}) = y$ has two roots near $t = y$, with $t/y = 1 \pm \sqrt{-2K\lambda_k/X} + O(X^{-1})$. Here $K = \gamma_1 + ((\gamma-1)^2 + \zeta(2) + 1)/2 > 0$. We give the next coefficients, a convergent two-generator transseries, the leading coefficient of every exponential sector, and a moving-fold normal form with local branch interchange. Finite residue formulas determine all chart coefficients from finite Taylor data. Proofs, executable exact checks, independent floating-point comparisons, an audit of the inspected repository material, and further research questions are included.

Status. The theorems are proved in this manuscript, but this is an unrefereed research contribution, not a Lean formalization. Classical continuation, Gamma expansions, implicit-function arguments, and analytic fold mechanisms are credited rather than claimed as new. The model-specific resonance classification, explicit critical-locus identities, and sector formulas are the principal candidate contributions; publication priority has not been established by an exhaustive literature search.

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1 Problem, contribution, and relation to the repository

The two selected repository areas provide complementary ingredients. The polylogarithm project organizes exact identities, parameter derivatives, Gamma and polygamma relations, and Stieltjes constants. The transseries project develops support-controlled inversion, analytic realizations, critical charts, and exponential sectors. Its critical-Hahn article already uses polylogarithms with a fixed nonintegral order to build a convergent multivariable inverse [6, 7, 8]. We do not claim that polylogarithmic inversion or the analytic implicit-function method is absent from the repository.

The question addressed here is narrower and different: what happens when *the order itself approaches an integer at the same time as the argument approaches one*? Two effects must be kept separate. At an integer, a Gamma pole cancels one zeta-series pole and produces logarithms. At certain complex values of the scaled order, the leading logarithmic coefficient cancels again. That second cancellation changes the inverse scale and, for the dilogarithm, creates a two-sheet chart.

1.1 A representative result

Let γ_1 use the standard Laurent convention

$$\zeta(1+z) = \frac{1}{z} + \sum_{r\geq 0} \frac{(-1)^r \gamma_r}{r!} z^r, \qquad \gamma_0 = \gamma. \tag{1.1}$$

Fix $k \in \mathbb{Z} \setminus \{0\}$ and put $\lambda_k = 2\pi i k$. Define

$$K = \gamma_1 + \frac{(\gamma-1)^2 + \zeta(2) + 1}{2}, \qquad b = 1 - \gamma - \frac{4K}{3}. \tag{1.2}$$

For large positive X, the two roots described above satisfy

$$t_\pm = e^{-X} \left\{ 1 \pm \sqrt{\frac{-2K\lambda_k}{X}} + \frac{b\lambda_k}{X} + O_k(X^{-3/2}) \right\}. \tag{1.3}$$

Both roots use the branch of $\operatorname{Log} t$ near the positive real axis; the two signs do not come from assigning different logarithm branches to the same input. They are two distinct solutions of one locally specified analytic equation. At exactly integer order the small positive inverse instead has size e^{-X}/X. Thus even the leading inverse scale is not uniform in a complex neighborhood of the order.

The conclusions of this paper concern genuine convergent multivariable Taylor or square-root charts. They do not assert Borel summability, resurgence, or Stokes constants. Local branch interchange at a fold and monodromy around $t = 0$ are explicitly distinguished.

1.2 What is inherited and what is developed

Jonquière's continuation formula is the starting identity [1, Eq. 25.12.12]. The Gamma Taylor series and the zeta Laurent convention are classical [2, 3]. The mechanism by which a simple critical point gives a square root is classical, and is already used extensively in the transseries collection. The contribution here is not a new version of the implicit-function theorem. It is the explicit choice of a pole-cancelled family, the two complementary order scalings, the Gamma–zeta critical-locus formulas, the exceptional integer-order classification, and the arithmetic coefficients of the resulting inverse sectors.

The reading and search scope is recorded in Section 10 and `provenance/SOURCE_AUDIT.md`. In particular, the supplied proofs do not assume the correctness of unreviewed repository arrivals.

2 A pole-cancelled polylogarithm family

For $n \geq 1$ define

$$\mathcal{F}_{n,\varepsilon}(t) = (-1)^n n! \left\{ \mathrm{Li}_{n+1+\varepsilon}(e^{-t}) - \sum_{j=0}^{n-1} \zeta(n+1+\varepsilon-j) \frac{(-t)^j}{j!} \right\}. \tag{2.1}$$

We work initially on $0 < |t| < 2\pi$, $|\arg t| < \pi$, with the branch continued from positive real t. Write $L = -\operatorname{Log} t$, and set

$$g_n(z) = \Gamma(1-z) \prod_{j=1}^{n} (1+z/j)^{-1}, \tag{2.2}$$

$$Z(z) = \zeta(1+z) - z^{-1}, \qquad Z(0) = \gamma, \tag{2.3}$$

$$R_n(t,z) = n! \sum_{j \geq 0} \frac{(-1)^{j+1} \zeta(z-j)}{(n+j+1)!} t^j. \tag{2.4}$$

Let $H_n^{(r)} = \sum_{j=1}^{n} j^{-r}$ and $H_n = H_n^{(1)}$. When $n = 0$ is used in a ladder formula, set $H_0^{(r)} = 0$, $g_0(z) = \Gamma(1-z)$, and interpret empty sums and products in the usual way. The same normal-form and jet formulas then apply to $\mathcal{F}_{0,\varepsilon}(t) = \mathrm{Li}_{1+\varepsilon}(e^{-t})$.

Theorem 2.1 (Exact holomorphic cancellation)**.** *For $|\varepsilon| < 1$ and the stated t-branch,*

$$\mathcal{F}_{n,\varepsilon}(t) = t^n \mathcal{B}_n(\varepsilon, L) + t^{n+1} R_n(t, \varepsilon), \tag{2.5}$$

$$\mathcal{B}_n(\varepsilon, L) = \frac{1 - g_n(\varepsilon) e^{-\varepsilon L}}{\varepsilon} + Z(\varepsilon). \tag{2.6}$$

The quotient in (2.6) is removable at $\varepsilon = 0$, and

$$\mathcal{B}_n(0, L) = L + H_n, \qquad R_n(0, \varepsilon) = -\frac{\zeta(\varepsilon)}{n+1}, \qquad R_n(0,0) = \frac{1}{2(n+1)}. \tag{2.7}$$

The series for R_n converges normally on compact subsets of $\{|t| < 2\pi,\ |\varepsilon| < 1\}$. All fixed order derivatives in ε may be taken term by term there.

Proof. For noninteger s, the classical continuation formula gives

$$\mathrm{Li}_s(e^{-t}) = \Gamma(1-s) t^{s-1} + \sum_{j \geq 0} \zeta(s-j) \frac{(-t)^j}{j!}.$$

At $s = n+1+\varepsilon$, the Gamma functional equation yields

$$(-1)^n n!\, \Gamma(-n-\varepsilon) = -\frac{g_n(\varepsilon)}{\varepsilon}.$$

After subtracting the first n Taylor terms, the term with $j = n$ is $t^n \zeta(1+\varepsilon)$. Combining it with the Gamma term gives (2.6); reindexing the remaining series gives (2.4). Since $g_n(0) = 1$, its numerator vanishes at $\varepsilon = 0$. Also $g_n'(0) = \gamma - H_n$, so its limiting value is $L - (\gamma - H_n) + \gamma = L + H_n$.

For completeness, normal convergence does not follow merely from a pointwise cancellation. On a compact set $|\varepsilon| \le r < 1$, the zeta functional equation [4] and the uniform Gamma-ratio estimate give, for $j \ge 2$,

$$\left|\frac{\zeta(\varepsilon - j)}{(n+j+1)!}\right| \le C_r (2\pi)^{-j} j^{-n-1+r}.$$

The two initial terms are holomorphic in the parameter disk. This bound proves normal convergence when $|t| \le \rho < 2\pi$. Cauchy's formula in a slightly larger parameter disk then justifies all fixed parameter derivatives. The equality at $\varepsilon = 0$ follows by removable continuation. □

Proposition 2.2 (Differential ladder and positive integral)**.** *With* $\mathcal{F}_{0,\varepsilon}(t) = \mathrm{Li}_{1+\varepsilon}(e^{-t})$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{F}_{n,\varepsilon}(t) = n\mathcal{F}_{n-1,\varepsilon}(t), \qquad \mathcal{F}_{n,\varepsilon}(t) = n\int_0^t (t-u)^{n-1}\,\mathrm{Li}_{1+\varepsilon}(e^{-u})\,\mathrm{d}u. \tag{2.8}$$

The integral holds for positive t *and* $|\varepsilon| < 1$*. For real* ε *in this range,* $\mathcal{F}_{n,\varepsilon}$ *is positive and strictly increasing on* $t > 0$*. Consequently every sufficiently small positive target has a unique positive inverse.*

Proof. Differentiate (2.1), using $\mathrm{d}\,\mathrm{Li}_s(e^{-t})/\,\mathrm{d}t = -\,\mathrm{Li}_{s-1}(e^{-t})$, and cancel the differentiated Taylor polynomial. The constants at $t = 0$ vanish for $n \ge 1$ by (2.5), since $\Re\varepsilon > -1$. Repeated integration gives the second identity. Local integrability of the integrand follows from its u^{ε} or logarithmic singularity. For real ε, the defining series of $\mathrm{Li}_{1+\varepsilon}(e^{-u})$ consists of strictly positive terms. The integral and the differential ladder give positivity and strict monotonicity, including $n = 1$. □

The subtraction in (2.1) is essential. For $n \ge 2$, inversion of the raw quantity $\zeta(n+1) - \mathrm{Li}_{n+1}(e^{-t})$ is ordinarily governed by a nonzero linear term, not by $t^n \,\mathrm{Log}(1/t)$. This paper concerns a normalized Taylor remainder, not an incorrectly assigned singular leading term for the unsubtracted polylogarithm.

3 All integer-order jets and their exact monodromy

The Gamma expansion gives

$$\log g_n(z) = (\gamma - H_n)z + \sum_{j\ge 2} \frac{\zeta(j) + (-1)^j H_n^{(j)}}{j} z^j, \qquad |z| < 1. \tag{3.1}$$

Define $a_n = \gamma - H_n$ and $b_{n,j} = \zeta(j) + (-1)^j H_n^{(j)}$.

Theorem 3.1 (Complete logarithmic jet)**.** *For every* $r \ge 0$*,*

$$\partial_\varepsilon^r \mathcal{F}_{n,\varepsilon}(t)\big|_{\varepsilon=0} = t^n P_{n,r}(L) + t^{n+1} R_{n,r}(t), \tag{3.2}$$

$$P_{n,r}(L) = (-1)^r \gamma_r - r!\,[z^{r+1}] \exp\left((a_n - L)z + \sum_{j\ge 2} \frac{b_{n,j}}{j} z^j\right), \tag{3.3}$$

$$R_{n,r}(t) = n! \sum_{j\ge 0} \frac{(-1)^{j+1}\zeta^{(r)}(-j)}{(n+j+1)!} t^j. \tag{3.4}$$

In particular $\deg_L P_{n,r} = r+1$*, with leading coefficient* $(-1)^r/(r+1)$*. Every coefficient in* (3.3) *is a finite polynomial in harmonic numbers, ordinary zeta values, Euler's constant, and one Stieltjes constant.*

Proof. Expand the removable quotient (2.6). Its coefficient of ε^r is minus the coefficient of z^{r+1} in $g_n(z)e^{-zL}$, plus $(-1)^r\gamma_r/r!$. Equation (3.1) yields (3.3); normal convergence proves (3.4). The highest power of L can come only from e^{-zL} and is $-r!(-L)^{r+1}/(r+1)! = (-1)^r L^{r+1}/(r+1)$. □

The first three polynomials are

$$P_{n,0}(L) = L + H_n, \tag{3.5}$$

$$P_{n,1}(L) = -\gamma_1 - \frac{(L-a_n)^2 + b_{n,2}}{2}, \tag{3.6}$$

$$P_{n,2}(L) = \gamma_2 + \frac{(L-a_n)^3}{3} + b_{n,2}(L-a_n) - \frac{2b_{n,3}}{3}. \tag{3.7}$$

The ladder also gives the exact finite-polynomial relation

$$nP_{n,r}(L) - P'_{n,r}(L) = nP_{n-1,r}(L). \tag{3.8}$$

This can be proved either by differentiation of (3.2), or from $g_{n-1}(z) = (1+z/n)g_n(z)$. Its analytic tails satisfy the corresponding coefficient identity.

Proposition 3.2 (Sharp unipotent monodromy of every jet)**.** *Let M be counterclockwise continuation once around $t = 0$ within the logarithmic covering of $0 < |t| < 2\pi$, and set $\Delta = M - 1$. Then*

$$(M-1)\mathcal{F}_{n,\varepsilon}(t) = -t^{n+\varepsilon} g_n(\varepsilon)\frac{e^{2\pi i\varepsilon} - 1}{\varepsilon}. \tag{3.9}$$

If $J_{n,r}(t) = \partial_\varepsilon^r \mathcal{F}_{n,\varepsilon}(t)\big|_0$, then

$$\Delta^{r+2} J_{n,r} = 0, \qquad \Delta^{r+1} J_{n,r} = -r!(2\pi i)^{r+1} t^n \neq 0. \tag{3.10}$$

Proof. Only t^ε changes in (2.5); this proves (3.9). For the jet, continuation sends L to $L - 2\pi i$ and leaves the analytic tail unchanged. A finite difference lowers the degree of a polynomial by one. The $(r+1)$st difference of its leading term is $(-1)^r r!(-2\pi i)^{r+1} = -r!(2\pi i)^{r+1}$. □

This is a direct consequence of classical continuation, not a claim of a new polylogarithm monodromy theorem. Its useful feature here is that the nilpotence index and top finite difference are exact, independent of n, and compatible with every order derivative and normalized remainder.

4 The primary, uniform inverse chart

Let

$$\mathcal{A}(\lambda) = \frac{1 - e^{-\lambda}}{\lambda}, \quad \mathcal{A}(0) = 1, \qquad \lambda_k = 2\pi i k \quad (k \in \mathbb{Z} \setminus \{0\}). \tag{4.1}$$

The zeros of $\mathcal{A}$ are exactly the λ_k, and each is simple. On the real line $\mathcal{A}(\lambda) = \int_0^1 e^{-u\lambda}\,du > 0$. For a positive target y put

$$X = \frac{1}{n}\log(1/y), \quad v = X^{-1}, \quad h = X^{-1}\log X, \quad \varepsilon = \lambda/X, \quad \tau = e^{-X}\{X\mathcal{A}(\lambda)\}^{-1/n}. \tag{4.2}$$

For complex λ choose $\log\mathcal{A}$ on a simply connected domain avoiding its zeros; the root in (4.2) uses that choice. The logarithmic sheet is specified by $\operatorname{Log} t = -X - (\log X + \log\mathcal{A}(\lambda))/n + \log U$. The theorem below is on this sheet; it is on the initial branch whenever its imaginary part lies in $(-\pi, \pi)$. If the parameter domain contains zero, normalize $\log\mathcal{A}(0) = 0$. On real parameter intervals we take the real logarithm of the positive function $\mathcal{A}$.

Theorem 4.1 (Convergent order-crossover inversion)**.** *Fix $n \geq 1$ and a compact subset $\mathscr{K}$ of such a parameter domain. There is a holomorphic function $U_n(v, h, \tau; \lambda)$ in a common polydisk around $(0, 0, 0)$, locally holomorphic in λ, such that $U_n(0, 0, 0; \lambda) = 1$ and*

$$t = \tau U_n(v, h, \tau; \lambda) \tag{4.3}$$

solves $\mathcal{F}_{n,\lambda/X}(t) = y$ after the substitutions (4.2). It is the unique solution with t/τ sufficiently close to 1, uniformly for $\lambda \in \mathscr{K}$. Its Taylor series is absolutely convergent after substitution and can be grouped as

$$U_n = \sum_{c \geq 0} \tau^c \sum_{N \geq 0} X^{-N} P_{N,c}(\log X; \lambda), \qquad \deg P_{N,c} \leq N. \tag{4.4}$$

The coefficients are holomorphic in λ, and $P_{0,c} = 0$ for $c \geq 1$. For real λ in a compact interval this is the unique small positive inverse.

Proof. With $t = \tau U$ we have

$$L = X + \frac{\log X + \log \mathcal{A}(\lambda)}{n} - \log U.$$

Define the holomorphic function

$$\Phi_n(U; v, h, \lambda) = \frac{1 - e^{-\lambda} g_n(\lambda v) \exp\{-\lambda h/n - \lambda v \log \mathcal{A}(\lambda)/n + \lambda v \log U\}}{\lambda} + v Z(\lambda v), \tag{4.5}$$

$$\mathcal{E}_n(U; v, h, \tau, \lambda) = U^n \{\Phi_n(U; v, h, \lambda) + v\tau U R_n(\tau U, \lambda v)\} - \mathcal{A}(\lambda). \tag{4.6}$$

At $\lambda = 0$ the quotient is removable as a function of *all* variables: its numerator vanishes identically when $\lambda = 0$. The limiting formula, which is sometimes computationally preferable, is

$$\Phi_n(U; v, h, 0) = 1 + h/n + v(H_n - \log U). \tag{4.7}$$

At $(v, h, \tau) = (0, 0, 0)$,

$$\mathcal{E}_n(U; 0, 0, 0, \lambda) = \mathcal{A}(\lambda)(U^n - 1), \qquad \partial_U \mathcal{E}_n(1; 0, 0, 0, \lambda) = n\mathcal{A}(\lambda) \neq 0.$$

The analytic implicit-function theorem gives the stated root. Uniqueness glues the local parameter charts. Compactness supplies common smaller radii and a common bound. Substituting the physical values into (4.6) is exactly (2.5) divided by y; hence this is an actual inverse, not only a formal solution.

Write its Taylor series as $\sum u_{abc}(\lambda) v^a h^b \tau^c$. After substitution its individual monomials become

$$u_{abc} X^{-a-b} (\log X)^b \tau^c.$$

This proves the degree assertion and absolute regroupability. When $v = 0$, equation (4.6) is independent of τ, so $U_n(0, h, \tau; \lambda)$ is as well. In particular $u_{00c} = 0$ for $c > 0$. For real parameters, conjugation and uniqueness make the root real and close to one; positivity and the uniqueness assertion in Proposition 2.2 identify it with the positive inverse. □

Corollary 4.2 (First crossover coefficient)**.** *Uniformly on the parameter compacts of Theorem 4.1,*

$$\frac{t}{\tau} = 1 - \frac{e^{-\lambda}\{(\log X + \log \mathcal{A}(\lambda))/n - a_n\} + \gamma}{n\mathcal{A}(\lambda) X} + O\left(\frac{(\log X)^2}{X^2} + \frac{|\tau|}{X}\right). \tag{4.8}$$

Proof. The terms of first degree in v, h in (4.5) are $e^{-\lambda} h/n + v\{e^{-\lambda}(\log \mathcal{A}/n - a_n) + \gamma\}$. Linearizing U^n at one gives (4.8). Holomorphy bounds the remaining slow terms quadratically. The difference between the full root and its value at $\tau = 0$ is divisible by $v\tau$ and is therefore $O(|\tau|/X)$ after substitution. □

4.1 Remainders and exponential resolution

If U_n is bounded by M on a closed polydisk of equal radius ρ, Cauchy's estimate gives an entirely explicit form of the truncation control. For $s = |v| + |h| + |\tau| < \rho$, truncation by total degree at most N obeys

$$|U_n - U_{n,\le N}| \le M \frac{(s/\rho)^{N+1}}{1 - s/\rho}. \tag{4.9}$$

Indeed, the sum of absolute monomials of degree d is at most s^d. The radii are existential here; no interval-certified numerical radius is claimed. Normal convergence is stronger than a bare Poincaré expansion.

A finite slow expansion has an algebraic error much larger than the first exponential sector. Therefore a claim about that sector must refer to the analytic coefficient of τ, or to subtraction of the entire convergent slow block, not to subtraction of a few powers of $1/X$. This distinction will also be crucial at the fold.

4.2 Comparison with the exact integer Lambert core

At $\varepsilon = 0$, ignoring only the analytic tail, the equation is $y = t^n(H_n - \log t)$. Its small positive core inverse is

$$t_{\mathrm{core}} = \left\{ \frac{-ny}{W_{-1}(-nye^{-nH_n})} \right\}^{1/n}. \tag{4.10}$$

To verify this, put $w = n(H_n - \log t)$. Then $we^{-w} = nye^{-nH_n}$, so $-w = W_{-1}(-nye^{-nH_n})$. The negative branch gives $w \to +\infty$ and hence $t \to 0$. This is a classical Lambert reduction [5], not a new special function identity. The primary chart recovers its logarithmic expansion, while retaining both order variation and the full analytic tail.

5 Secondary resonances and an exact Gamma–zeta critical locus

Fix $\lambda_k = 2\pi ik \neq 0$. Near this value it is no longer appropriate to divide the inverse scale by $\mathcal{A}(\lambda)$. Instead put

$$y = e^{-nX}, \qquad \eta = e^{-X}, \qquad t = \eta U, \qquad \lambda = \varepsilon X, \qquad \kappa = \frac{\lambda - \lambda_k}{\varepsilon}. \tag{5.1}$$

The last identity is used when $\varepsilon \neq 0$; the resulting local equation extends holomorphically to $\varepsilon = 0$. Since $e^{-\lambda} = e^{-\varepsilon\kappa}$, (2.5) becomes exactly

$$\mathcal{H}_n(U; \varepsilon, \kappa, \eta) = U^n \left\{ \frac{1 - g_n(\varepsilon) e^{\varepsilon(\log U - \kappa)}}{\varepsilon} + Z(\varepsilon) \right\} + \eta U^{n+1} R_n(\eta U, \varepsilon) - 1 = 0. \tag{5.2}$$

$$\mathcal{H}_n(U; 0, \kappa, \eta) = U^n(H_n + \kappa - \log U) + \eta U^{n+1} R_n(\eta U, 0) - 1. \tag{5.3}$$

Here $\log U$ is the local logarithm near a specified positive value of U. The variables $(\varepsilon, \kappa, \eta)$ are first independent analytic coordinates. The original problem is recovered by (5.1). Keeping this distinction is essential: $\eta = 0$ defines the analytic *singular core*, not a literal substitution into e^{-X} at finite X.

5.1 Critical points of the singular core at every order

Theorem 5.1 (Exact core critical locus)**.** *For each integer $n \ge 1$, the singular core $\mathcal{H}_n(U; \varepsilon, \kappa, 0)$ has a unique nearby critical solution $(U, \kappa) = (u_n(\varepsilon), k_n(\varepsilon))$ at which both the value and its U*

derivative vanish, where

$$u_n(\varepsilon) = \left\{ \frac{n+\varepsilon}{\varepsilon\zeta(1+\varepsilon)} \right\}^{1/n}, \tag{5.4}$$

$$k_n(\varepsilon) = \frac{\log g_n(\varepsilon) - \log n + (n+\varepsilon)\log u_n(\varepsilon)}{\varepsilon}. \tag{5.5}$$

The root and logarithms are chosen analytically near $u_n(0) = n^{1/n}$. *All displayed singularities at* $\varepsilon = 0$ *are removable, and*

$$k_n(0) = \kappa_n^* := \frac{1+\log n}{n} - H_n, \tag{5.6}$$

$$k_n(\varepsilon) = \kappa_n^* + K_n\varepsilon + O(\varepsilon^2), \tag{5.7}$$

$$K_n = \gamma_1 + \frac{(\gamma - 1/n)^2}{2} + \frac{\zeta(2) + H_n^{(2)}}{2} > 0. \tag{5.8}$$

At the core critical point, the following exact derivative identities hold:

$$\partial_\kappa \mathcal{H}_n = n, \qquad \partial_U^2 \mathcal{H}_n = -\frac{n(n+\varepsilon)}{u_n^2}, \qquad \partial_U \partial_\kappa \mathcal{H}_n = \frac{n(n+\varepsilon)}{u_n}. \tag{5.9}$$

Proof. For $\varepsilon \neq 0$, let $Q = g_n(\varepsilon)e^{-\varepsilon\kappa}$. The core equation and its derivative are

$$U^n\zeta(1+\varepsilon) - \frac{Q}{\varepsilon}U^{n+\varepsilon} = 1, \qquad nU^{n-1}\zeta(1+\varepsilon) - \frac{n+\varepsilon}{\varepsilon}QU^{n+\varepsilon-1} = 0.$$

Eliminating Q gives $U^n = (n+\varepsilon)/(\varepsilon\zeta(1+\varepsilon))$. The same equations give $QU^{n+\varepsilon} = n$, whose logarithm gives (5.5). Conversely these two identities imply both equations, so there is no lost condition. The analytic function $\varepsilon\zeta(1+\varepsilon)$ equals 1 at zero. This fixes the analytic root and logarithms in a sufficiently small disk. The numerator of (5.5) vanishes at zero, which proves removability.

To obtain its first coefficients, use (3.1) and

$$\log u_n(\varepsilon) = \frac{\log n}{n} + \left(\frac{1}{n^2} - \frac{\gamma}{n}\right)\varepsilon + \left(-\frac{1}{2n^3} + \frac{\gamma_1}{n} + \frac{\gamma^2}{2n}\right)\varepsilon^2 + O(\varepsilon^3).$$

Substitution gives (5.6)–(5.8). Direct differentiation, followed by $u_n^n\zeta(1+\varepsilon) = (n+\varepsilon)/\varepsilon$ and $Qu_n^{n+\varepsilon} = n$, gives (5.9). The Jacobian of $(\mathcal{H}_n, \partial_U\mathcal{H}_n)$ with respect to (κ, U) has determinant $-n^2(n+\varepsilon)/u_n^2 \neq 0$. The implicit-function theorem therefore proves local uniqueness, including at $\varepsilon = 0$.

We give a nonnumerical proof of the positivity in (5.8). Euler partial summation gives, first for $\Re s > 1$,

$$\zeta(s) = \frac{s}{s-1} - s\int_1^\infty \{x\}x^{-s-1}\,\mathrm{d}x.$$

The integral is holomorphic for $\Re s > 0$, and comparison at $s = 1$ gives

$$\gamma_1 = \int_1^\infty \{x\}(1-\log x)x^{-2}\,\mathrm{d}x > -\int_{\mathrm{e}}^\infty (\log x - 1)x^{-2}\,\mathrm{d}x = -\mathrm{e}^{-1}.$$

The strict inequality follows, for example, from the positive integral on $(1, \mathrm{e})$ and $0 \le \{x\} < 1$. Since $\zeta(2) > 1$ and $H_n^{(2)} \ge 1$, we obtain $K_n > 1 - \mathrm{e}^{-1} > 0$. □

Theorem 5.2 (First deformation by the full analytic tail)**.** *There are unique holomorphic critical functions $u_{n,c}(\varepsilon,\eta)$ and $\kappa_{n,c}(\varepsilon,\eta)$ near the core locus, satisfying $\mathcal{H}_n = \partial_U \mathcal{H}_n = 0$. Their expansions in η are*

$$\kappa_{n,c}(\varepsilon,\eta) = k_n(\varepsilon) + \eta\frac{\zeta(\varepsilon)u_n(\varepsilon)^{n+1}}{n(n+1)} + O(\eta^2), \tag{5.10}$$

$$u_{n,c}(\varepsilon,\eta) = u_n(\varepsilon) - \eta\frac{(1-\varepsilon)\zeta(\varepsilon)u_n(\varepsilon)^{n+2}}{n(n+1)(n+\varepsilon)} + O(\eta^2). \tag{5.11}$$

These remainders are uniform on smaller compact ε-disks.

Proof. The nonzero Jacobian just computed applies unchanged at $\eta = 0$, so the holomorphic implicit-function theorem gives both functions. Write $u_{n,c} = u_n + \eta u_1 + O(\eta^2)$ and $\kappa_{n,c} = k_n + \eta k_1 + O(\eta^2)$, and put $c = R_n(0,\varepsilon) = -\zeta(\varepsilon)/(n+1)$. The terms of first degree in η in the value and derivative equations are

$$nk_1 + cu_n^{n+1} = 0, \qquad -\frac{n(n+\varepsilon)}{u_n^2}u_1 + \frac{n(n+\varepsilon)}{u_n}k_1 + (n+1)cu_n^n = 0.$$

Solving this triangular system gives (5.10) and (5.11). Holomorphy proves the uniform remainder assertion. □

5.2 Why only the dilogarithm is an undetuned fold

Corollary 5.3 (Exceptional-order classification)**.** *Among $n \geq 1$, the limiting secondary equation at $\kappa = 0$ has a positive double root precisely for $n = 1$, when that root is $U = 1$. For every $n \geq 2$ it has exactly two positive roots, both simple. Each of those two simple roots extends to a holomorphic solution of the full secondary equation near $(\varepsilon,\kappa,\eta) = (0,0,0)$.*

Proof. For positive U, the function $f(U) = U^n(H_n - \log U)$ increases up to $U_* = e^{H_n - 1/n}$ and decreases thereafter. It vanishes at the limiting left endpoint and at $U = e^{H_n}$, and its maximum is e^{nH_n-1}/n. This maximum equals 1 at $n = 1$. For $n \geq 2$, $nH_n > 1 + \log n$, so the maximum is strictly greater than one. To see the strict inequality directly, $H_n \geq 1$ and $n > 1 + \log n$ for $n > 1$ already suffice. This proves the root count and simplicity. The last claim is the implicit-function theorem at each root. Equivalently, the critical detuning (5.6) is zero only at $n = 1$, and is negative for all larger n. □

The limiting roots also have a useful classical Lambert description. For real $\kappa > \kappa_n^*$ they are

$$U_j(\kappa) = \left\{\frac{-n}{W_j(-ne^{-n(H_n+\kappa)})}\right\}^{1/n}, \qquad j \in \{0,-1\}. \tag{5.12}$$

Indeed, substituting $w = n(H_n + \kappa - \log U)$ reduces the equation to $we^{-w} = ne^{-n(H_n+\kappa)}$. At $\kappa = \kappa_n^*$ the Lambert argument equals $-1/\mathrm{e}$ and the two branches meet. As real $\kappa \to +\infty$, the W_{-1} root satisfies $U \sim \kappa^{-1/n}$, whereas the W_0 root satisfies $U \sim e^{H_n+\kappa}$. These are identities for the limiting secondary core, not a proof of a global connection between every finite-X primary and secondary branch.

6 The dilogarithm: exact critical curve and a two-sheet chart

From now on $n = 1$, unless indicated otherwise. Set

$$a = \gamma - 1, \qquad K = \gamma_1 + \frac{a^2 + \zeta(2) + 1}{2}, \qquad b = -a - \frac{4K}{3}. \tag{6.1}$$

We suppress $n = 1$ in $\mathcal{H}$, u_c and κ_c.

6.1 A closed Gamma–zeta expression

Theorem 5.1 specializes to

$$u_1(\varepsilon) = \frac{1+\varepsilon}{\varepsilon\zeta(1+\varepsilon)}, \tag{6.2}$$

$$k_1(\varepsilon) = \log(1+\varepsilon) + \frac{\log\Gamma(1-\varepsilon) - (1+\varepsilon)\log(\varepsilon\zeta(1+\varepsilon))}{\varepsilon}. \tag{6.3}$$

The logarithms in (6.3) vanish at zero. Expanding gives

$$u_1(\varepsilon) = 1 - a\varepsilon + O(\varepsilon^2), \tag{6.4}$$

$$k_1(\varepsilon) = K\varepsilon + J\varepsilon^2 + O(\varepsilon^3), \tag{6.5}$$

$$J = -\frac{1}{2} + \frac{\zeta(3)}{3} + (1-\gamma)\gamma_1 + \frac{\gamma^2}{2} - \frac{\gamma^3}{3} - \frac{\gamma_2}{2}. \tag{6.6}$$

For example, $\log(\varepsilon\zeta(1+\varepsilon)) = \gamma\varepsilon - (\gamma_1 + \gamma^2/2)\varepsilon^2 + (\gamma_2/2 + \gamma\gamma_1 + \gamma^3/3)\varepsilon^3 + O(\varepsilon^4)$, which verifies every sign in (6.6) directly. The full critical curve satisfies

$$\kappa_c(\varepsilon,\eta) = K\varepsilon - \eta/4 + O_{\mathrm{tot}}(2), \qquad u_c(\varepsilon,\eta) = 1 - a\varepsilon + \eta/4 + O_{\mathrm{tot}}(2). \tag{6.7}$$

Here $O_{\mathrm{tot}}(d)$ denotes a convergent analytic series whose monomials have total degree at least d in the displayed small parameters.

6.2 Exact roots from analytic root moments

Theorem 6.1 (Convergent two-sheet normal form)**.** *There are holomorphic functions $C(\varepsilon,\kappa,\eta)$ and $D(\varepsilon,\kappa,\eta)$ near zero such that the full equation $\mathcal{H}(U;\varepsilon,\kappa,\eta) = 0$ has exactly two roots in a fixed sufficiently small disk about $U = 1$, counted with multiplicity, and they are exactly*

$$U_\pm = C(\varepsilon,\kappa,\eta) \pm \sqrt{D(\varepsilon,\kappa,\eta)}. \tag{6.8}$$

Their first analytic jets are

$$C = 1 + \frac{4}{3}\kappa + b\varepsilon + \frac{7}{12}\eta + O_{\mathrm{tot}}(2), \tag{6.9}$$

$$D = 2\kappa - 2K\varepsilon + \eta/2 + O_{\mathrm{tot}}(2). \tag{6.10}$$

Moreover

$$D = (\kappa - \kappa_c(\varepsilon,\eta))d(\varepsilon,\kappa,\eta), \qquad d(0,0,0) = 2, \tag{6.11}$$

where d is holomorphic and nonvanishing after shrinking the neighborhood. Thus a small loop in κ around its critical value, with the other parameters fixed, interchanges the two roots.

Proof. At zero parameters,

$$H_0(U) := \mathcal{H}(U;0,0,0) = U(1-\log U) - 1 = -\frac{(U-1)^2}{2} + O((U-1)^3).$$

Choose a circle Γ about 1 so small that this is its only zero in the closed disk. The function is nonzero on Γ. Holomorphy and Rouché's theorem show that the full equation has exactly two zeros inside, counted with multiplicity, for all sufficiently small parameters. Define

$$S_j = \frac{1}{2\pi i}\int_\Gamma U^j \frac{\partial_U \mathcal{H}}{\mathcal{H}}\,\mathrm{d}U \quad (j = 1,2), \qquad C = S_1/2, \qquad D = S_2/2 - C^2.$$

The contour is fixed and its denominator stays nonzero, so the moments are holomorphic even when the roots coalesce. If the two roots are u_+, u_-, the argument principle gives $S_j = u_+^j + u_-^j$. Hence $C = (u_+ + u_-)/2$ and $D = (u_+ - u_-)^2/4$, proving (6.8) without a prior choice of labels.

For the first jets put $\delta = U - 1$. Then

$$\frac{1}{H_0} = -2\delta^{-2} - \frac{2}{3}\delta^{-1} + O(1).$$

The first parameter-degree part Δ_1 of $\mathcal{H} - H_0$ is

$$\Delta_1 = \kappa U - \varepsilon U\{K + a\log U + (\log U)^2/2\} + \eta U^2/4.$$

Write $\Delta_1 = q_0 + q_1\delta + O(\delta^2)$, where $q_0 = \kappa - K\varepsilon + \eta/4$ and $q_1 = \kappa - (K+a)\varepsilon + \eta/2$. Expanding the logarithmic derivative and integrating by parts shows that the first variations are

$$S_1 - 2 = 2q_1 + \frac{2}{3}q_0, \qquad S_2 - 2 = 4q_1 + \frac{16}{3}q_0.$$

Consequently $C - 1 = q_1 + q_0/3$ and $D = 2q_0$ to first degree. These are (6.9)–(6.10). In particular $\partial_\kappa D(0) = 2$. A zero of D is precisely a double root, and its locus is the unique critical curve in Theorem 5.2. Division by $\kappa - \kappa_c$ therefore gives (6.11); for example the quotient is the integral of $\partial_\kappa D$ along the segment from κ_c to κ. The nonzero analytic factor has a local analytic square root. The remaining square root changes sign on one loop, which proves branch interchange. □

This local interchange is ordinary ramification of an analytic inverse. It is not a Stokes jump. Separately, Proposition 3.2 concerns continuation of the forward family around $t = 0$. Conflating these two operations would give incorrect branch bookkeeping.

6.3 A moving fold in the actual order parameter

Theorem 6.2 (Moving physical critical order)**.** *Fix $k \neq 0$ and write $v = X^{-1}$. There is a holomorphic function $\lambda_c(v, \eta)$ near $(0, 0)$ with $\lambda_c(0, 0) = \lambda_k$, characterized by*

$$\lambda_c - \lambda_k = \lambda_c v\, \kappa_c(\lambda_c v, \eta). \tag{6.12}$$

Its expansion is

$$\lambda_c = \lambda_k + K\lambda_k^2 v^2 - \frac{\lambda_k}{4} v\eta + O(v^3 + v^2\eta + v\eta^2). \tag{6.13}$$

On the physical specialization $\eta = e^{-X}$, set $\varepsilon_c = \lambda_c/X$ and $t_c = \eta u_c(\varepsilon_c, \eta)$. Then

$$\zeta(2 + \varepsilon_c) - \mathrm{Li}_{2+\varepsilon_c}(e^{-t_c}) = e^{-X}, \qquad \mathrm{Li}_{1+\varepsilon_c}(e^{-t_c}) = 0, \tag{6.14}$$

and this is a nondegenerate critical point of the inverse problem. In particular,

$$\varepsilon_c = \frac{\lambda_k}{X} + \frac{K\lambda_k^2}{X^3} + O_k(X^{-4} + e^{-X}X^{-2}), \tag{6.15}$$

$$t_c = e^{-X}\{1 + (1-\gamma)\lambda_k/X + O_k(X^{-2} + e^{-X})\}. \tag{6.16}$$

Proof. The left side of (6.12) minus its right side is holomorphic in (λ, v, η), vanishes at $(\lambda_k, 0, 0)$, and has λ derivative one there. The implicit-function theorem proves existence and uniqueness. Insert (6.7) and use $\lambda_c = \lambda_k + O(v^2 + v\eta)$ to obtain (6.13). For physical parameters, (5.1) makes the critical secondary equation identical to the original normalized equation. Its U derivative is the t derivative of $\mathcal{F}_{1,\varepsilon}(t)$, which equals $\mathrm{Li}_{1+\varepsilon}(e^{-t})$. Its second U derivative remains close to -1 and is nonzero. This proves (6.14) and nondegeneracy. Finally substitute (6.13) into (6.7) and divide by X where appropriate. □

The vanishing X^{-1} displacement in (6.13) is another expression of the exceptional-order classification. For general n, the same argument at the tuned critical detuning gives instead $\lambda_{n,c} = \lambda_k + \lambda_k \kappa_n^* v + O(v^2 + v\eta)$. This follows directly from (5.6); no analogous cancellation occurs for $n \geq 2$.

7 Convergent secondary transseries and every exponential sector

Fix the undetuned resonance $\lambda = \lambda_k$ and hence $\kappa = 0$. The small variables that preserve the fold splitting are

$$v = X^{-1}, \qquad w = \sqrt{v}, \qquad \eta = e^{-X}, \qquad r = \eta/v = Xe^{-X}, \qquad A_k = -2K\lambda_k \neq 0. \tag{7.1}$$

The letter A_k here is a nonzero constant, not the function $\mathcal{A}$.

Theorem 7.1 (Two-generator convergent inverse transseries)**.** *There are holomorphic functions $C_k(v, r)$ and $B_k(v, r)$ in a polydisk about zero such that, for all sufficiently large real X, the two roots near $t = e^{-X}$ are exactly*

$$t_\pm = e^{-X}\{C_k(X^{-1}, Xe^{-X}) \pm X^{-1/2} B_k(X^{-1}, Xe^{-X})\}. \tag{7.2}$$

One may select either square root of A_k, after which

$$C_k(0, r) = 1, \qquad B_k(0, r) = \sqrt{A_k + r/2}. \tag{7.3}$$

Consequently (7.2) is an absolutely convergent transseries on the finite positive monomial grid generated by $X^{-1/2}$ and Xe^{-X}, with the prefactor e^{-X}. Both roots use the same original logarithm branch, and they are distinct for large X.

Proof. Substitute $(\varepsilon, \kappa, \eta) = (\lambda_k v, 0, vr)$ into Theorem 6.1. Define

$$C_k(v, r) = C(\lambda_k v, 0, vr), \qquad T_k(v, r) = \frac{D(\lambda_k v, 0, vr)}{v}.$$

Every monomial in the numerator of the latter expression is divisible by v, because $D(0, 0, 0) = 0$. Thus T_k is holomorphic, not just bounded on a punctured set. Equation (6.10) gives $T_k(0, r) = A_k + r/2$. Since $A_k \neq 0$, shrink the polydisk and take its analytic square root B_k. This proves (7.3) and the exact formula. The functions $C_k(w^2, r) \pm wB_k(w^2, r)$ are holomorphic in (w, r), so their Taylor series converge absolutely on a smaller polydisk. Both generators tend to zero after (7.1); this proves convergence on the physical path. For each exponential action the powers of $X^{-1/2}$ have a lower bound, giving a well-based monomial support. Since $T_k \to A_k \neq 0$, the roots stay distinct. Finally $U_\pm \to 1$, so $t_\pm$ remain in a narrow sector about the positive real axis and share the logarithm specified at the start of the paper. □

A two-variable version of (4.9) gives a geometric error bound in $|w| + |r|$. More precisely, if the analytic root expression is bounded by M on a closed polydisk of radius ρ, then its Taylor polynomial of total degree at most N has error at most $M\theta^{N+1}/(1 - \theta)$, where $\theta = (|w| + |r|)/\rho < 1$. Multiplication by e^{-X} gives the error for t. The statement is local; the paper does not supply a maximal numerical polydisk or claim uniformity when k grows with X.

7.1 The first three algebraic corrections

Put

$$J_0 = \frac{\gamma_2}{2} - \frac{a^3}{6} - \frac{a(\zeta(2)+1)}{2} - \frac{\zeta(3)-1}{3}, \tag{7.4}$$

$$M_0 = \frac{a^2}{2} + (3a+1)K + \frac{13K^2}{9} + J_0. \tag{7.5}$$

(The subscript distinguishes M_0 from a Cauchy bound.)

Corollary 7.2 (Explicit Puiseux coefficients)**.** *At fixed $k \neq 0$ and with a fixed choice of $\sqrt{A_k}$,*

$$\frac{t_\pm}{e^{-X}} = 1 \pm \frac{\sqrt{A_k}}{X^{1/2}} + \frac{b\lambda_k}{X} \pm \frac{M_0\lambda_k^2}{\sqrt{A_k}\,X^{3/2}} + O_k(X^{-2}). \tag{7.6}$$

Proof. First set $\eta = 0$ and write $\delta = U - 1$. Give δ weight one and ε weight two. Expansion of (5.2) gives, through weight four,

$$\begin{aligned}\mathcal{H}(1+\delta;\varepsilon,0,0) = {} & -\delta^2/2 + \delta^3/6 - \delta^4/12 \\ & - \varepsilon\{K + (K+a)\delta + (a+1)\delta^2/2\} + \varepsilon^2 J_0 + O_{\mathrm{wt}}(5).\end{aligned}$$

Here the omitted analytic series has weight at least five. Substitute $\varepsilon = h^2$ and $\delta = ch + bh^2 + dh^3 + O(h^4)$. The coefficients of h^2, h^3, h^4 give successively $c^2 = -2K$, $b = -a - 4K/3$, and $cd = M_0$. This is the displayed formula with $\varepsilon = \lambda_k/X$. Convergence and the $O(h^4)$ error follow from Theorem 7.1 at $r = 0$, rather than from formal coefficient matching alone. For the full equation, analyticity in r shows that the additional change in U is $O(\sqrt{v}\,r + vr)$, hence $O(e^{-X}\sqrt{X})$ on the physical path. This is smaller than X^{-2} and proves (7.6) for the actual roots. □

7.2 The leading coefficient of every exponential action

The coefficients of the exponentially small sectors are most cleanly specified before substitution. Write

$$C_k(v,r) \pm \sqrt{v}\,B_k(v,r) = \sum_{m\geq 0} r^m U_{\pm,m}(v). \tag{7.7}$$

Each $U_{\pm,m}(w^2)$ is holomorphic in w. This independent-variable definition removes the ambiguity inherent in trying to read an exponential term beneath an unremoved algebraic error.

Theorem 7.3 (All-action leading law)**.** *For every fixed integer $m \geq 1$,*

$$U_{\pm,m}(v) = \pm\binom{1/2}{m} 2^{-m} A_k^{1/2-m}\sqrt{v} + O_{k,m}(v). \tag{7.8}$$

Thus the sector of $t_\pm$ of exponential action $m+1$, as defined by (7.7), has leading term

$$t_\pm^{[m]} = \pm\binom{1/2}{m} 2^{-m} A_k^{1/2-m} e^{-(m+1)X} X^{m-1/2}\{1 + O_{k,m}(X^{-1/2})\}. \tag{7.9}$$

Every one of these coefficients is nonzero. In particular, the first two additional sectors begin with

$$t_\pm^{[1]} \sim \pm\frac{e^{-2X}X^{1/2}}{4\sqrt{A_k}}, \qquad t_\pm^{[2]} \sim \mp\frac{e^{-3X}X^{3/2}}{32A_k^{3/2}}. \tag{7.10}$$

The full sector sum is absolutely convergent for all sufficiently large X; (7.9) itself is an asymptotic statement at fixed m.

Proof. At $v = 0$, C_k is independent of r, so $[r^m]C_k(v, r) = O(v)$ for $m \geq 1$. Equation (7.3) and Cauchy's formula in r give

$$[r^m]B_k(v, r) = \binom{1/2}{m} 2^{-m} A_k^{1/2-m} + O(v).$$

This proves (7.8). Now multiply by $e^{-X} r^m$ and use $r = Xe^{-X}$ to obtain (7.9). A half-integer binomial coefficient $\binom{1/2}{m}$ never vanishes, and $A_k \neq 0$; this proves nonvanishing. The binomial coefficients at $m = 1, 2$ give (7.10). Absolute convergence is the two-variable Taylor convergence already proved, with regrouping justified by absolute values. □

Corollary 7.4 (Analytic-tail universality of the leading actions)**.** *Replace $R_1(t, \varepsilon)$ by any holomorphic function $\widetilde{R}(t, \varepsilon)$ with $\widetilde{R}(0, 0) = c$, while leaving the singular block unchanged. All the local convergence statements remain valid. When $c \neq 0$, replace 2^{-m} by $(2c)^m$ in (7.8)–(7.9); all additional leading sectors remain nonzero. When $c = 0$, only the corresponding leading coefficient in (7.8) vanishes: the relative asymptotic formula (7.9) is not asserted, and a finer tail scaling may be needed.*

Proof. In the first-jet calculation the tail perturbation becomes $\eta c U^2$, so $D_\eta(0) = 2c$. Consequently $T_k(0, r) = A_k + 2cr$. The same analytic square root and coefficient argument apply without any other change. This proves the claim, including its qualification when $c = 0$. □

8 Finite residue formulas for every chart coefficient

The convergent existence theorems admit a finite coefficient algorithm. Although residue inversion is classical, the following formulas make the present three-variable charts directly computable and avoid an unbounded formal recursion for any fixed request.

Theorem 8.1 (Finite coefficient extraction for the simple chart)**.** *Fix n, λ in the setting of Theorem 4.1. Write $p = (v, h, \tau)$, $E_0(U) = \mathcal{A}(\lambda)(U^n - 1)$, and $Q(U, p) = (\mathcal{E}_n(U, p, \lambda) - E_0(U))/E_0(U)$. For every nonzero multi-index α of total degree N,*

$$[p^\alpha]U_n(p; \lambda) = \sum_{j=1}^{N} \frac{(-1)^j}{j} \operatorname{Res}_{U=1}[p^\alpha]Q(U, p)^j. \tag{8.1}$$

Each residue requires only finitely many Taylor coefficients of the explicit functions in (4.6).

Proof. Choose a small circle about $U = 1$ containing no other zero of E_0. For small p, $|Q| < 1$ on the circle, and the simple root inside is

$$U_n(p) = \frac{1}{2\pi i} \int U \frac{\partial_U \mathcal{E}_n}{\mathcal{E}_n} \, \mathrm{d}U = 1 - \frac{1}{2\pi i} \int \log(1 + Q) \, \mathrm{d}U.$$

For the last equality, integrate the derivative of $U \log(1 + Q)$ along the closed circle; this logarithm is single valued there by its convergent power series. Expanding the logarithm gives (8.1). Since Q has positive parameter degree, only $j \leq N$ can contribute. The pole order in the jth term is at most j, so a finite U-jet of its analytic numerator determines the residue. All expansions and integrals are uniformly convergent on the fixed contour for sufficiently small p, justifying coefficient extraction. □

Theorem 8.2 (Finite coefficient extraction at the double root)**.** *For the dilogarithm let* $p = (\varepsilon, \kappa, \eta)$*,* $H_0(U) = U(1 - \log U) - 1$*, and* $Q = (\mathcal{H} - H_0)/H_0$*. The root moments in Theorem 6.1 satisfy, for* $\alpha \neq 0$ *and* $N = |\alpha|$*,*

$$[p^\alpha] S_\ell = \ell \sum_{j=1}^{N} \frac{(-1)^j}{j} \operatorname{Res}_{U=1} U^{\ell-1} [p^\alpha] Q(U,p)^j, \qquad \ell = 1, 2. \tag{8.2}$$

Together with $C = S_1/2$ *and* $D = S_2/2 - C^2$*, this determines every coefficient of the double-root chart from finite Taylor data. Substitution, analytic square-root extraction, and ordinary convolution then determine every coefficient in Theorem 7.1.*

Proof. Use the same contour and logarithmic expansion as above, but start from $S_\ell = (2\pi i)^{-1} \int U^\ell \mathcal{H}_U / \mathcal{H} \, \mathrm{d}U$. Its base value is 2, because the base root at one has multiplicity two. Integration by parts introduces $-\ell U^{\ell-1}$, giving the stated formula. Only $j \leq N$ contributes. The pole order is now at most $2j$, which is still finite. Finally the analytic square root has nonzero constant term $\sqrt{A_k}$, so its coefficient recurrence divides only by $2\sqrt{A_k}$, never by a vanishing quantity. □

For the forward jets one can avoid even residues. If $g_n(z) = \sum_{m \geq 0} g_{n,m} z^m$ and $\log g_n(z) = \sum_{j \geq 1} \ell_{n,j} z^j$, then

$$g_{n,0} = 1, \qquad m g_{n,m} = \sum_{j=1}^{m} j \ell_{n,j} g_{n,m-j}, \tag{8.3}$$

where $\ell_{n,1} = a_n$ and $j\ell_{n,j} = b_{n,j}$ for $j \geq 2$. Multiplication by the finite jet of e^{-Lz} gives (3.3). This recurrence is obtained by differentiating $g_n = e^{\log g_n}$ and comparing coefficients, so it is exact over a symbolic coefficient ring. The verification program uses this recurrence rather than numerical fitting of constants.

9 Reproducible exact checks and numerical diagnostics

The accompanying computations have deliberately different evidentiary roles. The proofs above establish the analytic identities, convergence, and root counts. Exact symbolic computations check a finite set of coefficient identities. Floating-point evaluations test formulas and branch selections at specified sample points; they do not certify the last displayed digit, convergence radii, or publication novelty.

9.1 Exact algebra

The program `code/verify_exact.py`, using SymPy 1.14.0, completed **150 exact assertions**. Its scopes are: the leading logarithmic coefficient and exact monodromy degree for $0 \leq n, r \leq 4$; the differential jet ladder; the Gamma recurrence through degree six; the first root-moment coefficients of C, D; the Puiseux equation through fourth degree in $\sqrt{\varepsilon}$; the algebraic elimination and first-tail displacement at a general core critical point; the constants K, J; and the all-sector binomial convolution through degree eighteen.

Euler's constant, zeta values, and Stieltjes constants are independent symbols in these polynomial checks. No conjectured numerical relation between them is used. This exact algebra is neither a proof assistant formalization nor an independent verification of complex-analytic existence theorems. The complete receipt is `results/exact_checks.json`; the symbolic output is `results/symbolic_coefficients.json`.

9.2 Floating-point comparisons

The program `code/verify_numeric.py`, using mpmath 1.3.0 at **180 decimal digits**, completed **73 diagnostics**. It compares the pole-cancelled normal form with direct polylogarithm calls, including integer and complex orders; checks first order jets by a complex Cauchy coefficient calculation; solves primary inverses; checks the closed core critical formulas and their full-tail deformation; and checks both secondary roots and moving physical critical points against direct polylogarithms. Several comparisons share the same numerical library, so "independent" here means different evaluation routes, not independent software implementations or interval arithmetic.

Direct subtraction of two $O(1)$ values to recover a quantity of size e^{-X} costs approximately $X/\log 10$ decimal digits. The program therefore uses a large working precision and explicitly relaxes the direct residual threshold in its most ill-conditioned sample, $X = 320$. It records every tolerance and measured residual in `results/numeric_checks.json`. Finite tail truncations used in these numerical comparisons are diagnostics, not interval bounds for the omitted tail.

For the plus branch at $k = 1$, let $U^{(2)}$ retain the square-root and X^{-1} corrections in (7.6), and let $U^{(3)}$ also retain the $X^{-3/2}$ correction. The observed errors are

X	$\lvert U_+ - U^{(2)}\rvert$	$\lvert U_+ - U^{(3)}\rvert$	$X^{3/2}\lvert U_+ - U^{(2)}\rvert$
80	0.0425674	0.00810337	30.4587
160	0.0145921	0.00191344	29.5324
320	0.00503756	0.000458763	28.8367

These values are rounded diagnostics, not rigorously enclosed decimals. The constants used were approximately

$$K = 1.33902448494276296992829718554 33, \qquad b = -1.36258164482521682051090833747 35.$$

Their positivity or nonvanishing assertions in the theorems rely on exact arguments, not these decimal values.

9.3 How to replay

Run the following commands from the package directory:

```
python code/verify_exact.py
python code/verify_numeric.py --dps 180
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The scripts need only the pinned Python dependencies in `requirements.txt` and no network access after installation. The PDF build uses the included numerical table; replaying the numerical script regenerates it. The supplied `Makefile` separates mathematical replay from PDF compilation. No repository mutation is performed by these files.

10 Repository audit, corrections, and the novelty boundary

This is a targeted research audit, not a review of every statement in either large repository area. The observed `main` reference during the session was

`9ec0f0406afe7a5475b03ff65ed91f7e283979b8`.

Some initial index reads used the moving branch; later source and ledger reads were explicitly pinned to that commit. The source manifest records those distinctions and the returned blob identifiers rather than pretending that all reads formed one atomic checkout.

10.1 Overlap that must be credited

The critical-Hahn article [8] already constructs an actual convergent chart $x = TV(T, T^{2-\alpha})$ for a fixed exponent $1 < \alpha < 2$, with finite positive support generators, coefficient formulas, and a Cauchy remainder estimate. The inspected proof passage uses the analytic implicit-function theorem correctly for that fixed parameter regime. The present work does not supersede it and does not identify a defect in it. Our varying-order resonance scale and the exact Gamma–zeta critical locus are different statements. Uniformity through an integer endpoint cannot be inferred from a fixed-parameter radius; it requires the separate cancellation and scaling used here.

The transseries index also records existing moving-fold and inverse branch-gluing work. Accordingly, a square-root chart, root moments, Rouché's theorem, finite residue inversion, and local branch interchange are methods, not independent novelty claims of this paper. The specific resonance lattice, critical-detuning classification, closed critical locus, and the resulting sector coefficients are the candidate additions.

10.2 A historical overstatement already corrected in the ledger

The historical *Polylogarithm–Polygamma Bridge* abstract [10] uses stronger language about exact denominator classification and independent new constants than finite numerical searches establish. The source itself also distinguishes numerical independence evidence from proof. Crucially, the canonical editorial ledger [9] already directs replacement of the "exactly crystallographic" and independent-new-atom conclusions by exact reductions and bounded searches. This is therefore *not a newly discovered uncorrected theorem error*.

The integration proposal is to preserve that existing correction and, where standalone historical files are distributed, add an explicit supersession pointer to the canonical ledger. No numerical non-reduction should be promoted to a theorem of independence. This paper neither needs nor supplies any such independence theorem.

10.3 Preventive qualifications for the present contribution

Several tempting but incorrect simplifications are ruled out by the proofs. One must differentiate the pole-cancelled block, not two individually singular terms evaluated at the integer. The primary inverse must exclude the nonzero zeros of $\mathcal{A}$ or use a secondary chart. An analytic-core critical identity must not be labeled exact for the full polylogarithm without the tail deformation. Exponential coefficients must be extracted from analytic sector functions, not fitted below a fixed algebraic remainder. The two kinds of monodromy discussed above must remain separate. Finally, the positivity and small-real-inverse statements apply to the specified real family and do not exclude additional complex roots outside the local disks.

No new false mathematical assertion in the inspected *current canonical* material was established in this audit. Its limited scope would not justify claiming that the remaining corpus is error-free. The paper supplies self-contained proofs for the new results so that their validity does not depend on accepting unaudited sources.

10.4 Publication-priority limitations

A bounded web search for varying-order polylogarithm inversion, Gamma–zeta resonance, Stieltjes/fold combinations, and polylogarithm critical loci did not establish an exact antecedent for the combined formulas here. Most relevant verified public background was classical continuation and Lambert inversion in the DLMF. This limited search is not a bibliographic proof of originality. In particular, local fold methods are widespread, and equivalent formulas may occur in different notation. Independent mathematical review and a broader literature comparison are required before making a publication-priority claim.

11 Further research questions and concrete next steps

11.1 Matching the primary and secondary charts

A local primary chart excludes λ_k, while the secondary chart keeps $\kappa = (\lambda - \lambda_k)/\varepsilon$ bounded. Construct an overlapping atlas when $1 \ll |\kappa| \ll X$, with a uniform remainder and an explicit identification of which secondary root continues the small primary inverse along a specified path. Formula (5.12) suggests the matching law but does not prove the finite-X analytic continuation. The path dependence and possible encounters with the moving fold should be part of the statement, not hidden in a branch convention.

11.2 A growing resonance index

All secondary theorems here fix k. Determine the largest regime $k = k(X)$ for which a common rescaling and quantitative error remain valid. The constraint $|\varepsilon| = 2\pi|k|/X < 1$ is necessary for the present parameter disk, but it is not by itself a uniform root-separation or convergence estimate. A useful target is a theorem uniform when $|k| = o(X)$ with explicit dependence on $K\lambda_k$ and the nearest singularities in the order variable.

11.3 Higher criticality by controlled Taylor subtraction

The natural family yields a nondegenerate fold at its tuned critical point, since (5.9) never vanishes near zero. Add finitely many adjustable analytic counterterms to force the first several U derivatives to vanish. Can one construct an equally explicit Gamma–zeta parametrization of a cubic or higher critical locus, together with a convergent multi-sheet chart and all-action leading coefficients? The nondegeneracy and parameter-rank conditions must be proved before assigning a formal multiplicity.

11.4 Lerch and arithmetic twists

Replace the Riemann polylogarithm by a Lerch-type series with a rational shift or periodic weights. The local singular amplitude and regular Laurent coefficients then involve Hurwitz zeta and generalized Stieltjes constants. Determine precisely which periodic components preserve the resonance lattice and which cancel its leading residue. An attractive goal is a finite Fourier criterion for the appearance, disappearance, or splitting of the secondary fold. No blanket extension is asserted without checking the new local singular block.

11.5 Vanishing analytic tails and delayed actions

Corollary 7.4 shows why the leading law changes when $c = 0$. If the first nonzero tail coefficient has higher order, classify the correct replacement for Xe^{-X}, locate the first new action, and compute the sector coefficients. The center term and the square-root discriminant may then compete on different grids; a complete classification should retain both, rather than only the square root.

11.6 Global zeros and inverse continuation

Equation (6.14) constructs local zeros of a polylogarithm whose order tends to one, together with a prescribed adjacent-order value. Investigate whether these local critical points lie on distinct global critical curves, how those curves meet the usual argument cuts, and whether their inverse monodromies generate additional relations. The local two-root count in this paper says nothing about all other roots of the equation.

11.7 Certified radii and formalization

Produce interval-certified contour bounds for $\mathcal{H}$ and $\mathcal{E}_n$ on explicit disks, yielding numerical convergence radii and root enclosures rather than floating residuals. A sensible formalization order is: the finite coefficient algebra; the cancellation identity from a formalized continuation formula; normal convergence and order differentiation; the analytic root-count lemma; and the final substitutions. The supplied manifest separates these dependencies. Existing transseries Lean modules provide related support and inversion foundations, but this package contains no Lean declarations and claims no new formal coverage.

12 Conclusion

Integer-order pole cancellation is a natural interface between the exact polylogarithm calculus and convergent transseries inversion. Keeping that cancellation intact reveals a second, complex resonance lattice invisible to a fixed-order expansion. The singular core has a closed critical locus in Gamma and zeta functions at every order. The first full-tail correction is explicit, and the dilogarithm is exactly the order at which the undetuned secondary scaling already has a double inverse root. Its convergent square-root chart produces a complete leading law for every exponential sector. These results are local and model-specific, but their proofs, coefficient formulas, branch conventions, and replay artifacts make them suitable for statement-level review and integration.

References

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